Robotics and Computer Integrated Manufacturing 59 (2019) 143–157

Contents lists available at ScienceDirect

Robotics and Computer Integrated Manufacturing

journal homepage: www.elsevier.com/locate/rcim

Full length Article

Multi-objective optimization for energy-efficient flexible job shop scheduling problem with transportation constraints

Dai Min[a,*], Tang Dunbing[b], Giret Adriana[c], Salido Miguel A.[c]

[a] College of Mechanical Engineering, Yangzhou University, Yangzhou 225127, China
[b] College of Mechanical and Electrical Engineering, Nanjing University of Aeronautics and Astronautics, Nanjing 210016, China
[c] Departamento de Sistemas Informáticos y Computación/AI2, Universitat Politècnica de València, Camino de Vera s/n 46022, Valencia, Spain

ARTICLE INFO

Keywords:
Flexible job shop scheduling problems
Multi-objective optimization
Energy consumption
Genetic algorithm
Transportation time

ABSTRACT

Manufacturing enterprises nowadays face huge environmental challenges because of energy consumption and associated environmental impacts. One of the effective strategies to reduce energy consumption is by employing intelligent scheduling techniques. Production scheduling can have significant impact on energy saving in manufacturing system from the operation management point of view. Resource flexibility and complex constraints in flexible manufacturing system make production scheduling a complicated nonlinear programming problem. To this end, a multi-objective optimization model with the objective of minimizing energy consumption and makespan is formulated for a flexible job shop scheduling problem with transportation constraints. Then, an enhanced genetic algorithm is developed to solve the problem. Finally, comprehensive experiments are carried out to evaluate the performance of the proposed model and algorithm. The experimental results revealed that the proposed model and algorithm can solve the problem effectively and efficiently. This may provide a basis for the decision makers to consider energy-efficient scheduling in flexible manufacturing system.

1. Introduction

Manufacturing enterprise is facing a strong economic pressure as well as huge environmental challenges because of energy consumption and associated environmental impacts. According to the international energy outlook 2016 reference case, the industrial sector consumed 54% of global delivered energy in 2012, and its energy consumption grows by an average of 1.2% per year from 2012 to 2040 [1]. In China, manufacturing companies consumed more than 50% of the national electricity energy for production and generated at least 26% of the total carbon dioxide emission [2]. Thus, it is significant to explore energy-efficient mechanisms to reduce energy consumption for saving energy costs and supporting environment-friendly manufacturing activities [3–5].

Research on reducing energy consumption of manufacturing processes has mostly concentrated on the machine level, the product level, and the production management level. Due to the considerable capital investment, it may not be appropriate for small- and medium-size enterprises to design energy-saving machine tools and develop energy-efficient products in the perspective of the machine level and product level. On the other hand, increasing interests have been attracted to investigate production scheduling incorporated with energy efficiency issues in terms of production management methodologies. The existing research has largely focused on energy-efficient scheduling problems with regard to production operations. However, the transportation operations of jobs between machine tools have not been taken into account properly. In the field of production scheduling, each job of the schedule needs to be transported to the next machine tool by transportation resources such as Automated Guided Vehicle (AGV) after the completion of the job on a previous machine tool. In fact, there are strong interactions between production and transportation operations in real-world industries. On the one hand, the production operation (e.g. different machine tools' selection of two successive operations for a job) can affect transportation time of the AGV. On the other hand, the transportation time could result in different waiting times of machine tools in terms of different operation sequences. Furthermore, the transportation operations of transportation resources generate an amount of energy consumption due to the lack of scheduling optimization [5–7]. From the production management perspective, optimization of both the job shop scheduling and the AGVs scheduling in an integrated way can be considered an efficient strategy for manufacturing companies to decrease energy consumption. Therefore, in this paper, the transportation times between machine tools will be incorporated into the production scheduling to study the total energy

* Corresponding author.
E-mail address: daimin@yzu.edu.cn (M. Dai).

https://doi.org/10.1016/j.rcim.2019.04.006
Received 7 October 2018; Received in revised form 4 April 2019; Accepted 5 April 2019
Available online 10 April 2019
0736-5845/ © 2019 Elsevier Ltd. All rights reserved.

consumption. In addition, according to the number of AGVs, there are two types of transportation systems: single-AGV and multi-AGV transportation system [8]. Here the multi-AGV system is considered to perform the transportation of the jobs from one machine tool to another.

As it is well known, different kinds of shop floor environments have been discussed in the literature with regard to energy efficiency. In this paper, an energy-efficient flexible job shop scheduling problem (EFJSP) with transportation constraint is investigated for the following reasons. First, studies on the EFJSP have almost concerned the related energy consumption of the production operation but ignoring the transportation energy consumption. This study considers both the production and transportation operations to meet the requirements of practical cases better, and this research is to further enrich the energy-related scheduling theory for decision makers. Second, to perform different performance criteria in practical situations, further improvement is required on the optimization algorithms to make them more effective and efficient to optimize multi-objective functions by developing meta-heuristic approaches. Accordingly, the motivation of the research is to establish an energy-efficient multi-objective flexible job shop scheduling model with transportation constraint. Due to its complexity, a new multi-objective genetic algorithm is developed and comprehensive experiments are implemented to evaluate the performance of the proposed algorithm.

The outline of this paper is organized as follows. In Section 2, the related literature is reported. In Section 3, a multi-objective optimization model for the EFJSP is formulated. A meta-heuristic approach is devised to solve the optimization problem in Section 4. In Section 5, the proposed model and algorithm are evaluated. In Section 6, the conclusions and future work are presented.

2. Related work

There have been increasing studies on energy-related scheduling such as energy-efficient scheduling, low-carbon scheduling, and sustainable scheduling in different shop floor environments in the literature [9–11]. Research on reducing environmental impacts (like energy consumption and carbon emission) of production scheduling has mainly focused on three kinds of shop floors: the single machine scheduling environment, the flow shop scheduling environment, and the job shop scheduling environment.

From the single machine scheduling perspective, Mouzon et al. devised a mathematical programming model for optimizing the scheduling problem of a single CNC machine that considered the energy consumption and total completion time [12]. Afterwards, Mouzon and Yildirim discussed a multi-objective optimization schedule that minimized the total energy consumption and the total tardiness on a machine and employed a greedy randomized adaptive search algorithm to solve it [13]. Additionally, in order to optimize the total energy consumption and maximum completion time in a single machine environment, a mathematical model based on a multi-objective genetic algorithm was proposed by Mouzon and Yildirim [14]. Shrouf et al. modeled energy consumption costs for a single machine during production processes and used genetic algorithm to obtain optimal solutions [15]. In another work, Che et al. formulated a mixed-inter linear programming model to minimize both the total energy consumption and maximum tardiness for a single machine scheduling problem, and an advanced ε-constraint method was devised to obtain the Pareto front of the problem [16].

From the flow shop scheduling perspective, Fang et al. established new mixed integer linear programming formulations that considered simultaneously the peak total power consumption, the associated carbon footprint and the makespan for a flow-shop scheduling problem [17,18]. Ding et al. explored the permutation flow shop scheduling problem with the total carbon emissions and the makespan as objectives and considered two new multi-objective optimization algorithms for solving the problem [19]. In another study, Mansouri et al. introduced a two-machine flow shop scheduling optimization problem with the energy consumption and makespan and used a novel heuristic approach for the problem [20]. Moreover, Lu et al. described a mathematical model considering both energy consumption and makespan for a permutation flow shop scheduling problem and proposed a hybrid multi-objective backtracking search algorithm to solve the problem [21]. In addition to the classical energy-efficient flow shop scheduling, there have been reports on hybrid (or flexible) flow shop scheduling for energy efficiency. Bruzzone et al. investigated an energy-aware scheduling problem for a given flexible flow shop and designed a mixed integer programming formulation and an original meta-heuristic algorithm to realize energy savings [22]. Luo et al. defined a new hybrid flow shop scheduling problem with the objective of maximizing both production and energy efficiency and provided an ant colony optimization method to minimize makespan as well as electric power cost [23]. Yan et al. proposed an energy-efficient flexible flow shop model with the total energy consumption and makespan and developed an improved nesting optimization algorithm for the multi-level optimization model [24]. In addition, Dai et al. built an energy-efficient scheduling model of a flexible flow shop and adopted a genetic-simulated annealing algorithm to optimize the multi-objective problem with the energy consumption and makespan [25].

From the job shop scheduling perspective, Liu et al. presented a bi-objective scheduling problem that minimized total electricity consumption and total weighted tardiness for a classical job shop environment and developed new multi-objective genetic algorithms to obtain the solution [2,26]. Afterwards, May et al. analyzed a production scheduling problem with the objectives of energy consumption and makespan in a job shop system and designed a green genetic algorithm to find Pareto front solutions [27]. Zhang and Chiong studied the job shop scheduling problem with the total weighted tardiness as well as the total energy consumption and devised a hybrid genetic algorithm to solve the bi-objective optimization problem [28]. Additionally, in order to optimize both energy consumption and makespan, Salido et al. evaluated an extended job shop scheduling problem with different processing speeds of machines and used a genetic algorithm to solve it [29].

It is noted that flexible job shop scheduling problem is a general form of the classical job shop scheduling, involving operation flexibility, sequencing flexibility and processing flexibility, and is more practical in real-world applications. As flexible job shop scheduling problems are complex and non-deterministic polynomial-time (NP-hard), most of them cannot be solved by exact techniques. Thus, intelligent algorithms like genetic algorithm, particle swarm optimization, ant colony optimization, artificial immune algorithm, and hybrid optimization approaches are commonly used to solve these problems.

As a whole, there are relatively few researchers discussing energy-efficient scheduling especially for the FJSP. Mokhtari and Hasani investigated an energy-efficient FJSP with three objective functions, i.e., the total completion time, total availability of the system and energy consumption, and devised an enhanced evolutionary algorithm to cope with the optimization problem [30]. In another study presented by Liu et al. [31], a multi-objective optimization model with the objective of minimizing product carbon footprints and makespan in a flexible job shop scheduling environment was analyzed in detail, and a hybrid fruit fly optimization algorithm was designed to solve it. Yin et al. proposed a low-carbon scheduling model aimed at optimizing productivity, energy consumption and noise reduction by employing a multi-objective genetic algorithm for the FJSP [32]. In addition, Zhang et al. formulated a new mixed-integer linear mathematical model for an energy-efficient flexible job shop scheduling problem and discovered more energy-efficient dispatching rules via an efficient gene expression programming algorithm [33]. In another recent work, Wu and Sun considered an energy consumption model of machines at different operating states for the FJSP and developed a green scheduling heuristic algorithm to optimize the makespan, the energy consumption and the

numbers of turning-on/off machines [34]. Wang et al. proposed an operation-based integrated chart to characterize the dynamic characteristics of machining processes for energy-saving FJSP and employed a two-staged optimization method to optimize machine selection and operation sequence [35]. Gong et al. established a multi-objective optimization mathematic model considering processing time, green production and human factor for a double FJSP and designed a new hybrid genetic algorithm to solve the problem [36].

Based on the above literature review, most of energy-efficient scheduling problems just considered energy consumption yielded by production operations. However, transportation operation of AGVs has a non-negligible contribution to the total energy consumption that was ignored in previous works. Nouri et al. pointed out that the job shop scheduling problem (JSP) with transportation resources is a generalization of the classical JSP consisting of two sub-problems: the job shop problem and the generic vehicle scheduling problem [8]. Rossi and Dini proposed a promising approach to the scheduling of flexible manufacturing systems including alternative machine tools, setup and transportation times [37]. Zhang et al. introduced job shop scheduling problems for makespan objective minimization with transportation constraints [38]. Afterwards, Liou and Hsieh discussed job transportation times for the multi-stage flow shop group scheduling problem with the objective of makespan optimization [39]. Additionally, Karimi et al. addressed the FJSP with transportation times and devised a hybrid imperialist competitive algorithm to minimize the maximum makespan [40]. As a relatively related research, an energy-efficient scheduling with controllable transportation time was investigated in a permutation flow shop environment [21].

As it can be inferred from the above works, the scheduling objective in most of them is just completion time measures. However, the energy-efficiency criterion has not been taken into account, while, as mentioned before, production and transportation operations are major sources of energy consumption. Therefore, we have designed a flexible job shop scheduling problem with transportation constraint in which not only the traditional objective function like makespan is considered but also energy consumption is modeled as another objective function. As the multi-objective flexible job shop scheduling problem is NP-hard, an enhanced genetic algorithm (EGA) is proposed to find the optimized schedules.

3. Multi-objective optimization model of the EFJSP

3.1. Problem description

Formally, an energy-efficient flexible job shop scheduling problem (EFJSP) with transportation constraint can be defined as follows. There exist a set of n jobs $J = \{J_1, J_2, ...,J_n\}$ and a set of m machine tools $M = \{M_1, M_2, ...,M_m\}$. Each job $J_i$is composed of N_i operations $(O_{i1}, O_{i2}, ...,O_{iN_i})$ to be processed sequentially. Each operation O_{ij} can be processed by one of a subset of eligible machine tools $M_{ij} \subseteq M$. That is, the same operation O_{ij}can be processed by different machine tools, which consumes different processing times and different energy consumptions. In addition, when a job J_i has finished an operation O_{ij}on a machine tool M_w, an AGV is required to convey the job move from machine tool M_w to machine tool M_k. The AGV also consumes an amount of energy during the transportation time. The problem is to simultaneously determine the machine tool selection for each operation and the operation sequence on each machine tool to satisfy two efficiency criteria: schedule efficiency that is measured in terms of makespan and energy efficiency that is calculated by the total energy required for a schedule. Additionally, the required assumptions for EFJSP are shown in Table 1.

To make it easier to understand, an instance of the EFJSP is given in Table 2. As it can be observed, there are 3 jobs to be processed on 5 machine tools. Job J_1, J_2, J_3 have 3, 2, 3 operations, respectively. Each operation can be processed by one or more machine tools. For each machine tool, Table 2 shows the processing time and corresponding total energy consumption for each operation. For instance, the first operation O_{11} of job J_1 can be processed on machine tools M_1 and M_3. The processing time on M_1 and M_3 is 6 min and 9 min, respectively; the corresponding total energy consumption on M_1 and M_3 is 2170 kWh and 1460 kWh, respectively. After completing the preceding operation of a job on a machine tool, an AGV will convey the job from the current machine tool to another machine tool. The transportation times between different machine tools are shown in Table 3.

3.2. Notations

The notations used in the flexible job shop, energy consumption, transportation resource and throughout the paper are as follows:

Flexible job shop	
Indexes	
i, u, i^-	Index of jobs
j, v, j^-	Index of operations
k, w	Index of machine tools
q	Index of position
Sets	
J	Set of jobs, $J = \{J_1, J_2, ...,J_n\}$
M	Set of machine tools, $M = \{M_1, M_2, ...,M_m\}$
M_{ij}	Set of available machine tools for operation O_{ij}, $M_{ij} \subseteq M$
Constants	
m	Number of machine tools
n	Number of jobs
m_{ij}	Number of available machine tools for operation O_{ij}
N_i	Number of operations for job J_i
Q_k	Number of operations processed on machine tool M_k
M	A big positive number
Variables	
C_i	Completion time of job J_i
$C_{\max}$	Completion time of the last job
S_{ij}	Starting time of operation O_{ij}
T_{ij}	Processing time of operation O_{ij}
B_{kq}	Starting time of the operation allocated to the qth position on machine tool M_k
S_{ijk}	Starting time of operation O_{ij} on machine tool M_k
C_{ijk}	Completion time of operation O_{ij} on machine tool M_k
$C_{i^-j^-k}$	Completion time of the preceding operation of operation O_{ij} on machine tool M_k
T_{ijk}	Processing time of operation O_{ij} on machine tool M_k
t_{jk}	Setup time of machine tool M_k for the jth operation
X_{ijk}	Assignment binary variable that is set to 1 if operation O_{ij} is to be processed on machine tool M_k, and 0 otherwise
Y_{ijkq}	Sequencing binary variable that is set to 1 if operation O_{ij} is to be processed in qth position on machine tool M_k, and 0 otherwise
Energy consumption	
Parameters	
PB_{jk}	Basic power of machine tool M_k for the jth operation
P_{ijk}	Cutting power of operation O_{ij} on machine tool M_k
P_k	Unload power of machine tool M_k
E	Total energy consumption for a schedule
SE	Setup energy consumption module
PE	Processing energy consumption module
IE	Idle energy consumption module
TE	Transportation energy consumption module
AE	Auxiliary energy consumption module
e	Auxiliary average energy requirement per unit time
Transportation resource	
Parameters	
TP	Transportation power of automatic guided vehicle (AGV)
$T_{i(j-1)jwk}$	Transportation time needed to move from machine tool M_w to machine tool M_k for two successive operations $O_{i(j-1)}$ and O_{ij} of job J_i

Table 1
The required assumptions for the EFJSP.

Number	Assumptions
1	All the jobs are independent, and are available at time zero.
2	All the machine tools are available at time zero.
3	There are no precedence relationships between operations of different jobs, but there are precedence relationships between different operations of the same job.
4	An operation can only select one machine tool for processing.
5	Each machine tool can process at most one operation at a time.
6	Preemptive operation is not allowed.
7	There are enough AGVs responsible for transportation operations, and all jobs can be transported by any of them.
8	Once completing the preceding operation of a job, the job can be transported immediately by an AGV.

3.3. Formulation of the EFJSP optimization model

As for the mathematical model of EFJSP, a mixed-integer linear programming model (MILP) is developed to minimize the makespan and energy consumption. The MILP model includes the objective function and constraints. To establish the objective function, the energy consumption modules of the production workshop are also given.

According to energy-consuming behaviors of each job, an operation processed on a machine tool consumes an amount of energy [25,41,42]. Moreover, if two successive operations of a job are executed on different machine tools, energy consumption yielded by moving the job between machine tools should be taken into account. Hence, the total energy consumption of a schedule is mainly composed of multiple energy consumption modules: energy consumption module in the setup state (*SE*), energy consumption module in the processing state (*PE*), energy consumption module in the idle state (*IE*), energy consumption module in the transportation state (*TE*), auxiliary energy consumption module in the entire production workshop (*AE*).

(1) Energy consumption module for the setup state (*SE*). *SE* is the energy required for machine preparation activities such as positioning, clamping, loading or unloading work pieces, changing tools and switching operations. It can be calculated by multiplying the basic power demand for maintaining the normal running of machine tool by the preparation time as shown in Eq. (1)

$$SE = \sum_{k=1}^{m} \sum_{j=1}^{O_k} PB_{jk} \cdot t_{jk} \tag{1}$$

(2) Energy consumption module for the processing state (*PE*). *PE* is the energy usage for material removal, which can be estimated by a polynomial fitting curve [42]. Hence the processing energy consumption is expressed by Eq. (2), in which α is the coefficient of the fitting curve.

$$PE = \sum_{k=1}^{m} P_k \cdot \left(\sum_{i=1}^{n} \sum_{j=1}^{N_i} T_{ijk} \cdot X_{ijk} \right) + \sum_{i=1}^{n} \sum_{j=1}^{N_i} \sum_{k=1}^{m} (1+\alpha) \cdot P_{ijk} \cdot T_{ijk} \cdot X_{ijk} \tag{2}$$

Table 3
The transportation times of AGVs between machine tools (unit: min).

Machine tool number	M_1	M_2	M_3	M_4	M_5
M_1	0	4	2	4	2
M_2	4	0	5	3	5
M_3	2	5	0	6	6
M_4	4	3	6	0	2
M_5	2	5	6	2	0

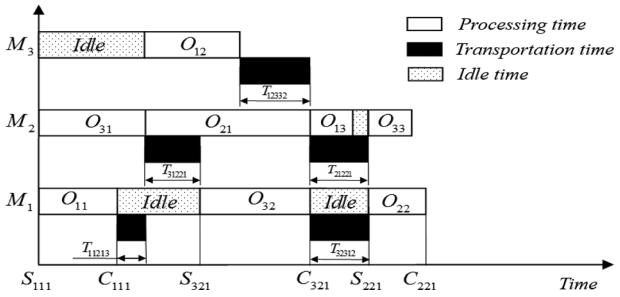


Fig. 1. The Gantt chart of the instance mentioned before.

(3) Energy consumption module for the idle state (*IE*). *IE* is the energy consumption for waiting to process next operation when a machine tool is at the idle running state. The idle time is the duration between two adjacent operations on the same machine tool, which is dependent on the machine tool selection, the operation sequence, and the related transportation time. Fig. 1 shows a Gantt chart of the instance mentioned before. For example, when machine tool M_1 is selected to process operations O_{11}, O_{32}, and O_{22}, the assignment binary variables X_{111}, X_{321}, X_{221} are set to 1 and two idle times are generated in terms of the operation sequence and transportation times. The total idle time of M_1 is calculated as C_{221}-((C_{111}-S_{111}) + (C_{321}-S_{321}) + (C_{221}-S_{221})). Similarly, the total idle time of M_2 *and* M_3 can be obtained. Hence, *IE* can be obtained by

Table 2
One instance of the EFJSP.

Job number	Operations	Machine tool candidates	(Processing time/min, Energy consumption/kWh)
J_1	O_{11}	M_1,M_3	(6,2170), (9,1460)
	O_{12}	M_1,M_2,M_3	(10,1340), (4,2730), (5,2600)
	O_{13}	M_1,M_2,M_3,M_4,M_5	(5,1610), (3,2640), (4,1730), (6,1500), (2,2790)
J_2	O_{21}	M_1,M_2,M_5	(7,1690), (10,1260), (4,1830)
	O_{22}	M_1,M_2,M_3,M_4	(4,2230), (8,1780), (6,1940), (5,2090)
J_3	O_{31}	M_2,M_5	(6,1780), (4,2700)
	O_{32}	M_1,M_2,M_3,M_4	(4,3230), (8,1800), (7,2640), (6,2730)
	O_{33}	M_1,M_2,M_5	(6,1460), (3,2230), (5,1690)

multiplying the unload power demand by the total idle time as shown in Eq. (3)

$$IE = \sum_{k=1}^{m} P_k \cdot (\max_{i,j} C_{ijk} - \sum_{i=1}^{n} \sum_{j=1}^{N_i} (C_{ijk} - S_{ijk}) \cdot X_{ijk}) \tag{3}$$

(4) Energy consumption module for the transportation state (*TE*). *TE* is the energy consumption of the transportation resource, such as automatic guided vehicle (AGV), for transporting a job whenever it requires transportation from one machine tool to another. The transportation time is dependent on different machine tools' selection of two successive operations for a job. For instance, if two successive operations O_{11} and O_{12} of job J_1 are assigned to M_1 and M_3, respectively, then assignment binary variables X_{111} and X_{123} are set to 1. Thus, the transportation time T_{11213} is 2 min according to Table 3. *TE* can be calculated by Eq. (4)

$$TE = \sum_{i=1}^{n} \sum_{j=2}^{N_i} \sum_{k=1}^{m} \sum_{w=1}^{m} TP \cdot T_{i(j-1)jwk} \cdot X_{i(j-1)w} \cdot X_{ijk} \tag{4}$$

(5) Auxiliary energy consumption module for the production workshop (*AE*). *AE* is the auxiliary energy required to support the production environment like lighting, heating, and air-conditioning, which can be obtained by Eq. (5)

$$AE = e \cdot \max_{1 \le i \le n} (C_i) \tag{5}$$

According to the problem description, the scheduling optimization objectives are to minimize the maximum completion time of all operations (namely, makespan) and the total energy consumption. Therefore, the MILP model of EFJSP requiring the minimization of the makespan and energy consumption is formulated as follows:

$$\min f1 = C_{\max} = \min(\max_{1 \le i \le n} (C_i)) \tag{6}$$

$$\begin{aligned} \min f2 &= E = SE + PE + IE + TE + AE \\ &= \sum_{k=1}^{m} P_k \cdot (\sum_{i=1}^{n} \sum_{j=1}^{N_i} T_{ijk} \cdot X_{ijk}) + \sum_{i=1}^{n} \sum_{j=1}^{N_i} \sum_{k=1}^{m} (1+\alpha) \cdot P_{ijk} \cdot T_{ijk} \cdot X_{ijk} \\ &\quad + \sum_{k=1}^{m} P_k \cdot (\max_{i,j} C_{ijk} - \sum_{i=1}^{n} \sum_{j=1}^{N_i} (C_{ijk} - S_{ijk}) \cdot X_{ijk}) \\ &\quad + \sum_{k=1}^{m} \sum_{j=1}^{O_k} PB_{jk} \cdot t_{jk} \\ &\quad + \sum_{i=1}^{n} \sum_{j=2}^{N_i} \sum_{k=1}^{m} \sum_{w=1}^{m} TP \cdot T_{i(j-1)jwk} \cdot X_{i(j-1)w} \cdot X_{ijk} + e \cdot \max_{1 \le i \le n} (C_i) \end{aligned} \tag{7}$$

subject to

$$\sum_{k=1}^{m} S_{ijk} \cdot X_{ijk} = S_{ij}; \quad i = 1, 2, \ldots, n; \ j = 1, 2, \ldots, N_i \tag{8}$$

$$\sum_{k=1}^{m} T_{ijk} \cdot X_{ijk} = T_{ij}; \quad i = 1, 2, \ldots, n; \ j = 1, 2, \ldots, N_i \tag{9}$$

$$S_{i(j-1)} + \sum_{k=1}^{m} \sum_{w=1}^{m} (T_{i(j-1)} + T_{i(j-1)jwk}) \cdot X_{i(j-1)w} \cdot X_{ijk} \le S_{ij}; \quad i = 1, 2, \ldots, n; \ j = 2, \ldots, N_i; \ k, w = 1, 2, \ldots, m \tag{10}$$

$$S_{ijk} \ge \max(C_{i(j-1)w} + T_{i(j-1)jwk}, C_{i^-j^-k}); \quad i, i^- = 1, 2, \ldots, n; \ j = 2, 3, \ldots, N_i; j^- = 1, 2, \ldots, N_{i^-}; k, w = 1, 2, \ldots, m \tag{11}$$

$$\sum_{k=1}^{m_{ij}} X_{ijk} = 1; \quad i = 1, 2, \ldots, n; \ j = 1, 2, \ldots, N_i \tag{12}$$

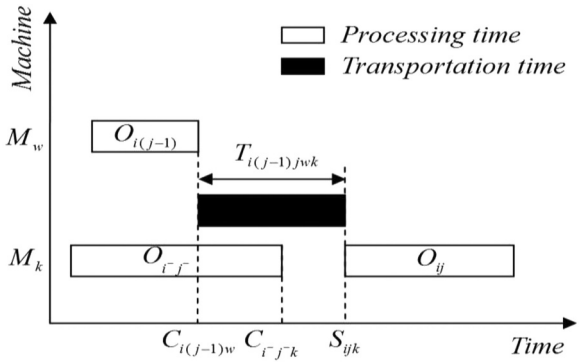


Fig. 2. The Gantt chart to illustrate the constraints.

$$\sum_{i=1}^{n} \sum_{j=1}^{N_i} Y_{ijkq} = 1; \quad k = 1, 2, \ldots, m; \ q = 1, 2, \ldots, Q_k \tag{13}$$

$$\sum_{q=1}^{Q_k} Y_{ijkq} = X_{ijk}; \quad i = 1, 2, \ldots, n; \ j = 1, 2, \ldots, N_i; \ k = 1, 2, \ldots, m \tag{14}$$

$$B_{kq} \le T_{ij} + \mathrm{M} \cdot (1 - Y_{ijkq}); \quad i = 1, 2, \ldots, n; \ j = 1, 2, \ldots, N_i; \ k = 1, 2, \ldots, m; \ q = 1, 2, \ldots, Q_k; \tag{15}$$

$$B_{kq} \ge T_{ij} - \mathrm{M} \cdot (1 - Y_{ijkq}); \quad i = 1, 2, \ldots, n; \ j = 1, 2, \ldots, N_i; \ k = 1, 2, \ldots, m; \ q = 1, 2, \ldots, Q_k; \tag{16}$$

The first objective (6) minimizes the makespan and the second objective (7) minimizes the total energy consumption. Constraints (8) and (9) address the starting time and processing time of each job after a machine tool is assigned. Constraint (10) means that there are precedence constraints between the operations of each job. In other words, the operation O_{ij} of job J_i should be started after the completion of the operation $O_{i(j-1)}$ of the job and the transportation before processing operation O_{ij}. Constraint (11) considers the relationship between the transportation constraint and the machine capacity constraint for two adjacent operations on the same machine tool. When the transportation time of the succeeding operation before processing is more than the completion time of the preceding operation, the machine capacity constraint set is contained in the transportation constraint set. Otherwise, the transportation constraint set is contained in the machine capacity constraint set. As shown in Fig. 2, there exist two adjacent operations ($O_{i^-j^-}$, O_{ij}) on machine tool M_k. Assuming that the completion time of the preceding operation $O_{i^-j^-}$ of operation O_{ij} on machine tool M_k is $C_{i^-j^-k}$, the completion time of the ($j-1$)th operation $O_{i(j-1)}$ of job J_i on machine tool M_w is $C_{i(j-1)w}$, and the transportation time needed to move from machine tool M_w to machine tool M_k for two successive operations of job J_i is $T_{i(j-1)jwk}$. If $C_{i^-j^-k}$ is later than $C_{i(j-1)w} + T_{i(j-1)jwk}$, the start time of the jth operation of job J_i on machine tool M_k should be after the transportation time of the operation $O_{i(j-1)}$. Otherwise, the start time of the jth operation of job J_i on machine tool M_k should be after the completion time of the operation $O_{i^-j^-}$. Constraint (12) guarantees that an operation can only select one machine tool for processing. Constraint (13) ensures that just one operation is assigned to one position on a machine tool. Constraint (14) presents a linking constraint between assignment variable X_{ijk} and sequencing variable Y_{ijkq}. Constraints (15) and (16) ensure linking constraints between start times B_{kq} and S_{ij}. In other words, if operation O_{ij} is processed in qth position on machine tool M_k, then $B_{kq} = S_{ij}$.

4. An enhanced genetic algorithm (EGA) for solving the EFJSP

In order to explore optimal or near-optimal solutions for the EFJSP, an enhanced meta-heuristic algorithm is developed based on a combination of genetic algorithm (GA), particle swarm optimization (PSO) and simulated annealing algorithm (SAA). GA is a powerful searching technique inspired by the natural evolutionary process, and it can quickly obtain good solutions in scheduling problems and other combinatorial problems. However, the major disadvantage of GA is that it can be trapped in a suboptimal solution owing to premature convergence. In order to overcome this drawback, the hybridization of GA with other meta-heuristics is an effective method. The working mechanism of PSO is, on the basis of animals' behavior of cluster activities, making the population movement from disorderly to orderly evolutionary computation in a complex solution space by sharing information from an individual. Due to its simplicity, quick convergence and high accuracy, PSO has been applied to solve optimization scheduling problems. The working mechanism of SAA is, on the basis of an algorithmic analogy with the annealing of materials, leading the material to a state corresponding to a minimum value of its internal energy. One of the major advantages of SAA is that it has an effective local search mechanism for accurately obtaining near solutions. Hence, this study proposes to incorporate the strengths of particle swarm optimization and simulated annealing algorithm into a genetic algorithm. Specifically, an information-sharing mechanism based on PSO is introduced to design the crossover operator of the GA, which improves the searching ability of the proposed algorithm. Then, a novel temperature update function of the SAA based on a Hill function is devised to replace the mutation operator of the GA, which moves to the better solution. The main components of the proposed enhanced genetic algorithm (EGA) are summarized as follows.

4.1. Representation

In the population, a multi-layer encoding rule is formulated for each individual, which contains information regarding the machine tool selection for each operation and the operation sequence on each machine tool. Each chromosome is comprised of two layer strings. Specially, the first layer encoding represents a machine gene-string and the second one represents an operation gene-string. The length of the machine gene-string equals to the length of the operation gene-string, and the total length (L) of a chromosome is $2 \times \sum_{i=1}^{n} N_i$, where N_i represents the number of operations for job J_i. For the first layer encoding, each gene is encoded in terms of the machine tools set of the corresponding job's operation. The gene values denote the index of the array of machine tool candidates set for each operation. For the second layer encoding, an operation-based representation is employed to address the operation gene-string. Different appearances of the same job number indicate different operations of the job and the pth appearance represents the pth operation of the job. In the EFJSP, after assigning machine tools and sequencing operations, a feasible schedule with respects to schedule efficiency (involving processing time, transportation time and idle time and so on) and energy efficiency (involving processing energy consumption, transportation energy consumption and idle energy consumption and so on) can be obtained. Hence, the information with regard to the schedule efficiency and energy efficiency needs to be taken into account in the decoding process. When each operation O_{ij} of a job selects a machine tool M_k for processing, the associated processing time and energy consumption regarding O_{ij} processed on M_k is considered. Further, in terms of the information that two successive operations of the job are assigned to machine tools, the transportation time between the machine tools is also considered.

Following the instance mentioned before (see Table 2), a chromosome (C) consisting of machine gene-string ($C1$) and operation gene-string ($C2$) is represented in Fig. 3. It can be observed that the length of the chromosome is $L = 16$ and both the length of machine gene-string and operation gene-string are equal to 8. For job J_1, it has three operations. Thus, the first three genes are assigned to J_1 in the machine gene-string ($C1$) of the chromosome. The value of each gene represents the corresponding machine tool selection of each operation for job J_1. For example, the 1st position '2' of the chromosome represents the second index of the array of machine tool candidates $\{M_1, M_3\}$ for operation O_{11}, which means that M_3 is selected to process the first operation of job J_1. Then, the associated processing time (9 min) and energy consumption (1460 kWh) regarding O_{11} processed on M_3 are considered (see Table 2); the 2nd position '3' of the chromosome represents the third index of the array of machine tool candidates $\{M_1, M_2, M_3\}$for operation O_{12}, which means that M_3 is selected to process the second operation of job J_1. Then, the associated processing time (5 min) and energy consumption (2600 kWh) regarding O_{12} processed on M_3 are considered. Since two successive operations O_{11} and O_{12} of job J_1 are assigned to the same machine tool M_3, the transportation time T_{11233} is 0 (see Table 3). According to the operation gene-string, a corresponding operation sequence can be determined. For instance,

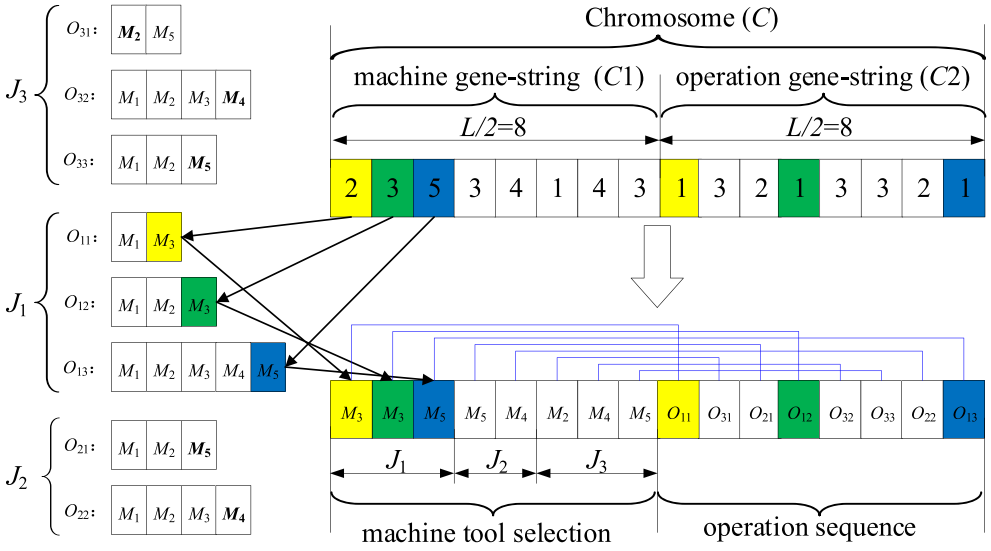


Fig. 3. A chromosome representation.

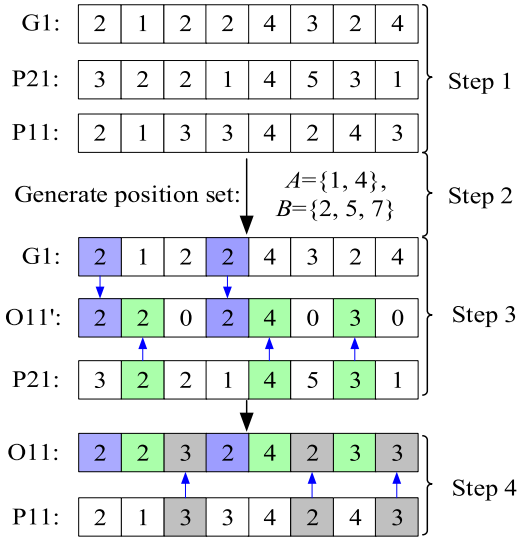


Fig. 4. Crossover operator for the machine gene-string.

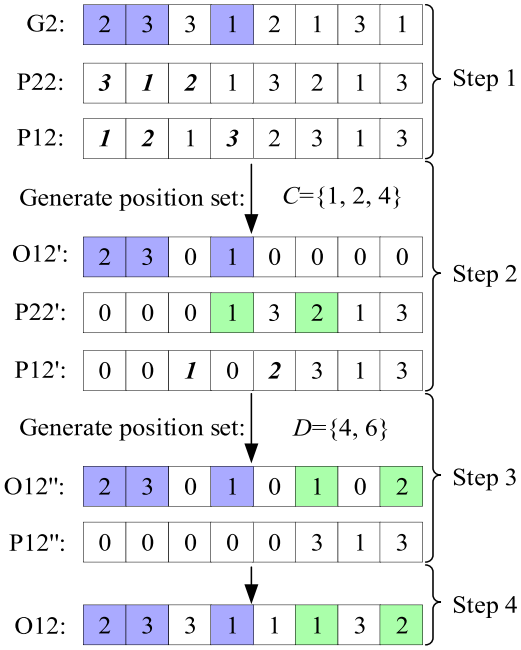


Fig. 5. Crossover operator for the operation gene-string.

number '1′ represents job J_1 and it has three different appearances in the operation gene-string (*C*2), which means job J_1 has three operations. The first appearance (in the 9th position) of number '1′ denotes the first operation of job J_1; the second one (in the 12th position) denotes the second operation and so on. Similarly, with regard to J_2 and J_3, the encoding and decoding operations can be also obtained. Therefore, an operation sequence with the assigned machine tools can be obtained: $O_{11}(M_3)$–$O_{31}(M_2)$–$O_{21}(M_5)$–$O_{12}(M_3)$–$O_{32}(M_4)$–$O_{33}(M_5)$–$O_{22}(M_4)$–$O_{13}(M_5)$.

4.2. Crossover operator

In this section, a new crossover operator inspired by the mechanism of PSO is adopted for EGA. On one hand, a guided individual is designed to assist the crossover operator. Based on the above-mentioned description, a set of chromosomes can be generated stochastically. According to the fitness evaluation of each chromosome, all the solutions are sorted by using the non-dominated sorting strategy, and then a crowding distance technique is employed to determine an order among chromosomes. Finally, the best individual is selected as the guided individual. On the other hand, two self-adaptive learning factors k_1 and k_2 are proposed for the crossover operator, where k_1 represents the self-adaptive learning factor toward the guided individual and k_2 represents the self-adaptive learning factor toward one parent individual.

The procedure of crossover operator for the machine gene-string is described as follows, and a crossover operator instance is shown in Fig. 4.

Step 1: Establish the guided individual (G1) and parent individuals (P11, P21) for the machine gene-string in terms of chromosomes.

Step 2: Select randomly $L/2 \times k_1$ integer gene positions in G1 to generate a position set *A*; select $L/2 \times k_2$ integer gene positions to generate a position set *B* except the gene positions of the set *A* in P21.

Step 3: Append the corresponding gene elements of the position set *A* in G1 to a new individual (O11') in order to generate a child individual, and retain in the same position in the new individual. Similarly, append the corresponding gene elements of the position set *B* in P21 to the new individual, and retain in the same position in the new individual. Here, "0″ represents an empty position in O11'.

Step 4: Check the remaining empty positions in O11', and orderly copy the corresponding gene elements of the empty positions in P11 into O11' so as to obtain the final offspring individual O11.

The procedure of crossover operator for the operation gene-string is described below, and a crossover operator instance is shown in Fig. 5.

Step 1: Determine a guided individual (G2) and parent individuals (P12, P22) for the operation gene-string in terms of chromosomes.

Step 2: Select randomly $L/2 \times k_1$ integer gene positions in G2 to generate a position set *C*; append the corresponding gene elements of the position set *C* in G2 to a new individual (O12') in order to generate a child individual, and retain in the same position in the new individual; read each corresponding gene element of the position set *C* in the parent individuals (P12, P22) from left to right, and replace the gene element with "0″ in (P12, P22) to generate new parent individuals (P12', P22') once it first appears. Note that "0″ represents an empty position.

Step 3: Select $L/2 \times k_2$ integer gene positions in the non-empty positions of P22' to generate a position set *D*; append sequentially the corresponding gene elements of the position set *D* in P22' to any empty position of O12' so as to yield another new individual O12''; read each corresponding gene element of the position set *D* in the new parent individual P12' from left to right, and replace the gene element with "0″ in P12' to generate another parent individual P12'' once it first appears.

Step 4: Sequentially append the non-empty gen elements of P12'' to the empty positions in O12'' to obtain the final offspring individual O12.

4.3. Mutation operator

GA can quickly obtain good solutions of combinatorial optimization problems, but a critical shortcoming is being liable to be trapped in a suboptimal solution, i.e. premature convergence. Fortunately, SAA has the ability to jump out of the suboptimal solution and search for a better solution [43]. Thus, an improved simulated annealing algorithm method is devised to replace the mutation operator of GA.

In this method, changing three key modules, i.e., the state generation module, acceptance probability module, and annealing rate module improves the performance of the algorithm. These modules are designed as follows:

(1) State generation function

State generation function is mainly used to obtain a new individual. For the machine gene-string, the new individual is generated as:

Step 1: Select randomly several positions in the parent individual.

Step 2: Change the values of these selected positions to other

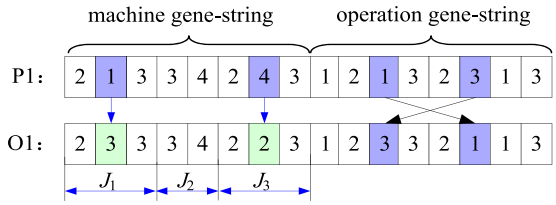


Fig. 6. A new individual based on state generation function.

machine numbers in the set of available machine tools of the corresponding operations.

For the operation gene-string, the new individual is generated as:

Step 1: Select two positions in the parent individual.

Step 2: Swap the corresponding gene elements of the selected positions.

Fig. 6 shows a new individual example based on state generation function.

(1) Acceptance probability module

A key feature of the SAA is the probability of acceptance when a new state function value is higher than the current one. To improve the performance of the SAA, an acceptance probability function based on Boltzmann function is defined as:

$$p(i \to j) = \begin{cases} 1, & F(i) \geq F(j) \\ \exp\left(\frac{F(i) - F(j)}{T}\right), & F(i) < F(j) \end{cases} \tag{17}$$

where $F(i)$ represents the new state function value; $F(j)$ represents the current function value; T is the current temperature.

(1) Annealing rate module

Annealing rate is another important module of the SAA. Annealing rate has a significant relation with the acceptance probability of inferior solutions, which can directly affect the speed and accuracy of the SAA. In general, the traditional annealing rate function is given by Eq. (18).

$$T(t) = k \times T(t-1) \tag{18}$$

where $k \in (0, 1)$ is an annealing rate coefficient. It usually ranges from 0.75 to 0.95; $T(t)$ and $T(t-1)$ are current annealing temperature and previous annealing temperature, respectively; t is the iteration number.

Due to the limitation of solution space, the SAA has the low acceptance probability of inferior solutions for Eq. (18) through many iterations, and it can hardly search for better individuals. The Hill function is a nonlinear function with non-negativity and monotonicity, it can adjust the range of solution space and has the ability to avoid premature convergence. Hence, a new annealing rate function inspired by the Hill function is proposed by Eq. (19) to enhance the search efficiency of the SAA.

$$T(t) = k \times \frac{T_0^n}{T_0^n + t^n} \tag{19}$$

where T_0 is a temperature threshold and it is related with an initial temperature of the SAA; n is the Hill coefficient.

4.4. The procedure of EGA

Based on the above the representation of crossover and mutation operators, the procedure of the proposed algorithm can be summarized with the following pseudo-code in Algorithm 1.

Algorithm 1
Enhanced genetic algorithm (EGA).

Begin
Initialize parameters, including population size (*NIND*), maximum number of generation (*Maxgen*), crossover probability (P_c), self-adaptive learning rate (k_1, k_2), Hill coefficient (n), Iteration number for the current temperature (L_n);
Initialize the population stochastically according the encoding scheme;
Determine a guided chromosome by evaluating each individual's objective value of performance criterion function;
While (*Maxgen* $> t$) **do**
$t: = t + 1$
Perform crossover operator:
While (*NIND* is not met) **do**
If ($P_c > rand$) **then**
Perform crossover operator for the machine gene-string with k_1 and k_2;
Perform crossover operator for the operation gene-string with k_1 and k_2;
End if
End while
Perform mutation operator:
While (Iteration number for the current temperature is not met) **do**
Generate a neighbor solution S_j from S_i according to the state generation function;
Evaluate the improvement of performance criterion function $\Delta = F(S_j) - F(S_i)$;
If ($\min[1, \exp(-\Delta/T)] > random[0, 1]$) **then**
$S_i = S_j$;
End if
End while
Update annealing rate function T by Eq. (19);
Update the best solution and the guided individual;
End while
Output best results;
End

5. Comprehensive experiments

In this section, the proposed mathematical model and EGA are tested using a real-world case for the EFJSP with the transportation time. First, the analysis of the algorithm parameter settings is conducted to investigate their impact on the performance of the EGA. Second, the EGA is employed to evaluate the energy-saving potential of the proposed model. Third, to demonstrate the effectiveness of the EGA, different scale instances with regard to the EFJSP are implemented by the proposed EGA versus the multi-objective genetic algorithm (MOGA). The comprehensive experiments were implemented by utilizing MATLAB programming language. The experimental tests were conducted on a personal computer with Intel Pentium (R) with 4 GB memory and 2.60 GHz processor.

5.1. Case introduction

For this study, a real-world case provided by a flexible workshop in Nanjing, China was tested. There are 10 types of jobs with 32 operations to be processed on 10 machine tools. There exist several automatic guided vehicles (AGVs) for transporting the jobs among machine tools, and the power demand of AGV is set to 3.5 kW. Thus, once completing an operation of a job, the job can be transported immediately by an available AGV. Additionally, the auxiliary power required to support the production environment is set to 1 kW. The processing times and setup times for all the operations on different machine tools are listed in Table 4, and the transportation times between different machine tools are given in Table 5. In addition, the related power for each machine tool is set in Table 6.

5.2. Sensitivity analysis of the algorithm parameters

In the EGA, the crossover operator with the PSO idea and the mutation operator with the SAA are two critical parts of the algorithm. The associated parameter settings may have significant influence on the performance of the EGA, including: the population size (*NIND*),

Table 4
Processing times for all the operations on different machine tools.

Number	Operations	Machine tools candidates	Processing time (min)	Setup time (min)
J_1	O_{11}	M_1,M_2,M_3,M_7	18,20,40,45	2,2,3,3
	O_{12}	M_9,M_{10}	20,14	3,2.5
	O_{13}	M_1,M_2,M_3,M_7	17,20,30,28	3,2,3.5,3
J_2	O_{21}	M_1,M_2,M_3,M_7	12,10,20,15	2,1.5,3,2
	O_{22}	M_1,M_2,M_8	15,20,30	2,2,3
	O_{23}	M_9,M_{10}	20,25	3,2.5
J_3	O_{31}	M_1,M_2,M_3,M_7	30,33,40,40	3,4,4,3.5
	O_{32}	M_9,M_{10}	20,15	2.5,3
J_4	O_{41}	M_1,M_2,M_3,M_7	11,10,10,25	1.5,1.5,2,3
	O_{42}	M_9,M_{10}	15,16	2,2
	O_{43}	M_1,M_2,M_3,M_7	15,17,29,30	2,2.5,4,4
	O_{44}	M_1,M_2,M_3,M_7	21,15,20,20	2.5,2,3,2
J_5	O_{51}	M_1,M_2,M_3,M_7	20,17,16,25	2,3,2.5,3
	O_{52}	M_9,M_{10}	20,20	3,4
	O_{53}	M_1,M_2,M_3,M_7	25,28,37,40	3,3,4,5
J_6	O_{61}	M_4,M_5,M_6	15,25,30	1.5,2,2
	O_{62}	M_4,M_5,M_6	25,20,30	3,2.5,4
	O_{63}	M_1,M_2,M_3,M_7	20,10,10,35	2,1,1,3
J_7	O_{71}	M_4,M_5,M_6	10,20,30	1.5,1.5,2
	O_{72}	M_4,M_5,M_6	15,15,20	2,3,2.5
	O_{73}	M_1,M_2,M_3,M_7	15,15,30,30	2,3,3,4
J_8	O_{81}	M_4,M_5,M_6	20,25,30	1.5,2,2,5
	O_{82}	M_1,M_2,M_3,M_7	15,20,30,40	2.5,3,3,4
J_9	O_{91}	M_4,M_5,M_6	10,18,23	1,2,2
	O_{92}	M_4,M_5,M_6	15,20,20	1.5,3,2.5
	O_{93}	M_1,M_2,M_3,M_7	5,5,7,11	1,1.5,2,2
	O_{94}	M_1,M_2,M_3,M_7	10,12,17,15	1.5,1.5,2,2.5
J_{10}	O_{101}	M_4,M_5,M_6	10,10,20	2,2.5,3
	O_{102}	M_4,M_5,M_6	15,10,25	1.5,2,2
	O_{103}	M_1,M_2,M_3,M_7	10,10,25,30	1,1.5,3,3.5
	O_{104}	M_1,M_2,M_3,M_7	20,15,30,40	3.5,2,2,4
	O_{105}	M_9,M_{10}	22,20	3,3

Table 5
Transportation times between different machines (unit: s).

Machine tool Number	M_1	M_2	M_3	M_4	M_5	M_6	M_7	M_8	M_9	M_{10}
M_1	0	152	170	193	100	112	173	165	142	133
M_2	152	0	131	138	152	169	120	170	162	143
M_3	170	131	0	160	151	140	132	171	122	140
M_4	193	138	160	0	140	140	165	170	140	198
M_5	100	152	151	140	0	103	102	170	180	192
M_6	112	169	140	140	103	0	142	140	148	150
M_7	173	120	132	165	102	142	0	150	162	160
M_8	165	170	171	170	170	140	150	0	141	120
M_9	142	162	122	140	180	148	162	141	0	153
M_{10}	133	143	140	198	192	150	160	120	153	0

maximum number of iterations (*Maxgen)*, crossover probability (P_c), learning rate (α), Hill coefficient (n) and iteration number for the current temperature (L_n). In order to investigate the effectiveness of the proposed algorithm, two other algorithms are implemented for comparison with the EGA in terms of these parameters. One is the new crossover operator with the PSO idea while keeping all the other parameters unaltered in the algorithm, which is called CGA; the other is the new mutation operator with the SAA while keeping all the other parameters unaltered in the algorithm, which is called MGA. Due to space limitations, a pilot experiment on the case with different parameters has been conducted in advance. The final parameter settings are summarized in Table 7.

Table 7
The parameter settings of EGA, CGA and MGA.

Parameter name	EGA	CGA	MGA
Population size (*NIND*)	100	100	100
Maximum number of generation (*Maxgen*)	500	500	500
Crossover probability (P_c)	0.8	0.8	0.8
Mutation rate (P_m)	–	0.1	–
Self-adaptive learning factors (k_1, k_2)	0.3, 0.3	0.3, 0.3	–
Annealing rate coefficient (n)	2	–	2
Iteration number for the current temperature (L_n)	10	–	10
Temperature threshold (T_0)	50	–	50

Table 8
The experimental results for EGA, CGA and MGA.

Algorithms	HV mean(std)	IGD mean(std)	Computation time (s) Mean
EGA	1.94(0.13)	1.96e−02(3.3e−03)	288.64
CGA	1.95(0.15)	4.41e−02(8.8e−03)	48.65
MGA	1.92(0.16)	1.97e−02(4.4e−03)	295.18

In the experiment, the hypervolume (HV) and inverted generational distance (IGD) are adopted as comprehensive indicators to evaluate the convergence and distribution of the algorithms. Larger HV and smaller IGD are more favorable for the performance of the algorithms. All the experimental tests were carried out 30 times independently. The experimental results with respect to the mean and standard deviation values of HV and IGD, as well as the computation time are shown in Table 8.

In Table 8, the EGA and CGA are superior to the MGA for the HV. The HV value of the EGA is slightly inferior to the HV value of the CGA, but there is no significant difference between them. The main reason is that the new crossover mechanism with the PSO idea in the EGA and CGA can improve the search efficiency by sharing the information of the guided individual and using the self-adaptive learning mechanism in the population. We can also observe that the EGA and MGA outperform the CGA for the IGD, and that the IGD value of the EGA is lowest among the algorithms. The new mutation mechanism with SAA in the algorithm can improve the comprehensive performance. Here, the new annealing rate function of the SAA plays a key role in the mutation operator.

Furthermore, to investigate the influence of different annealing rate functions on the acceptance probability of inferior solutions in the EGA, Fig. 7 shows the change trend of acceptance probability of inferior solutions for a traditional annealing rate function and new annealing rate function. As it can be observed, when k is set to 0.75 for the traditional annealing rate function, the acceptance probability decreases to 0.002 after running 36 iterations; when k is set to 0.85, the SAA reaches to the same acceptance probability after running 64 iterations; when k is set to 0.95, the SAA reaches to the same acceptance probability after running 202 iterations. For the new annealing rate function, we also conducted 36 iterations, 64 iterations and 202 iterations.

Table 6
Unload power of each machine tool.

Machine tool number	M_1	M_2	M_3	M_4	M_5	M_6	M_7	M_8	M_9	M_{10}
Rated power (kW)	20	15	6	12	10	5.5	7.5	3	5.5	10
Unload power (kW)	3.45	2.82	0.84	1.58	1.41	0.55	1.02	0.37	1.16	1.8

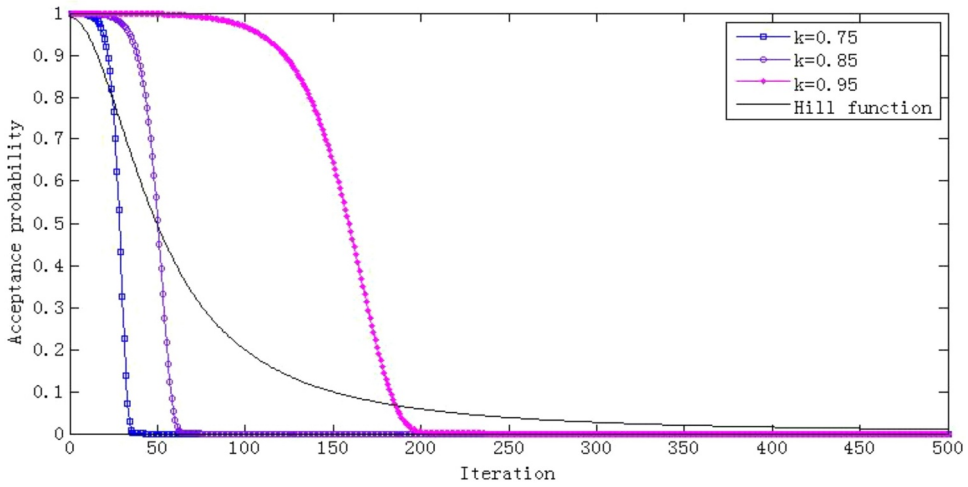


Fig. 7. The change of acceptance probability.

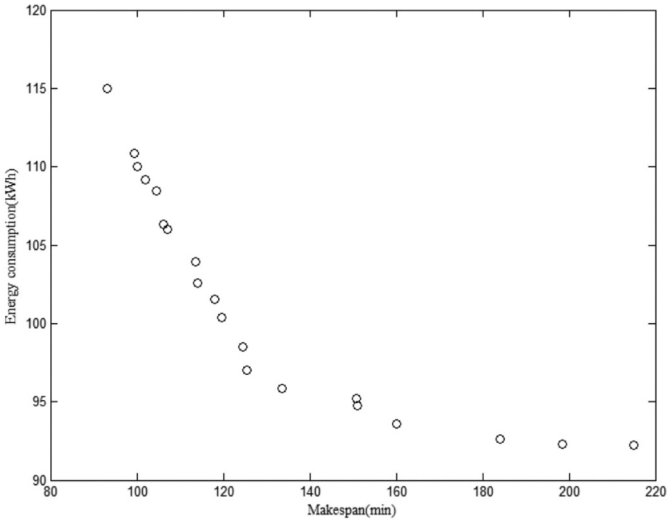


Fig. 8. The Pareto curve obtained by the proposed algorithm.

Table 9
The Pareto solutions of the case study.

Pareto solution numbers	Makespan (min)	Energy consumption (kWh)	evaluation matrix/ a_j
S1	104.53	108.46	0.4922
S2	184.07	92.65	0.7387
S3	93.03	114.96	0.3330
S4	124.53	98.49	0.7299
S5	118.07	101.52	0.6587
S6	133.70	96.29	0.7694
S7	102.03	109.20	0.4773
S8	107.07	105.99	0.5577
S9	113.50	103.94	0.6002
S10	106.23	106.30	0.5509
S11	215.07	92.21	0.6670
S12	198.50	92.29	0.7099
S13	100.07	110.00	0.4592
S14	151.07	94.74	0.7675
S15	159.30	94.25	0.7594
S16	99.53	110.86	0.4355
S17	**125.50**	**97.03**	**0.7701**
S18	114.00	102.59	0.6385
S19	119.68	100.36	0.6883
S20	150.83	95.22	0.7540

The acceptance probabilities of the inferior solutions are 0.652, 0.375, and 0.057, respectively. Compared to the traditional annealing rate function, the new annealing rate function has higher acceptance probability of inferior solutions.

In summary, the EGA is a promising approach. The integration of the new crossover operator and the new mutation operator has the more positive effect on the behavior of the proposed algorithm. In addition, the EGA is comparable with the other algorithms on the comprehensive performance, though it doesn't obtain the shortest computation time. The reason is that the traditional mutation has been replaced with the SAA that is an iterative improvement algorithm. The SAA is much more time-consuming than the mutation.

5.3. Evaluation of the energy-efficient scheduling model

To evaluate the energy-saving potential of the proposed model, the EGA is employed to find optimal or near-optimal solutions. The experimental results in terms of the maximum completion time and the total energy consumption obtained by the proposed algorithm are presented in Fig. 8, and the Pareto solutions are shown in Table 9. It can be observed that the solution space is quite large, in which the makespan ranges from 93.03 min to 215.07 min, the corresponding total energy consumption values are 114.96 kWh and 92.21 kWh. Based on the Pareto front presented by the black circle in Fig. 8, there exists a conflict between the two objective functions. If the makespan is minimized, the corresponding energy consumption gets worse values. When the makespan is at its minimum (93.03 min), the corresponding energy consumption is 114.96 kWh. When the makespan is sacrificed, the energy consumption can be reduced to the minimum value 92.21 kWh and the total energy consumption decreases to 24.67%.

Further, the decision makers can obtain the most satisfying energy-efficient schedule from the Pareto solutions according to the analytic hierarchy process [44]. First, according to the preference of decision makers, a relative comparison decision matrix D is established to compute the weight vector $W = (0.333\ 0.667)^{\mathrm{T}}$.

$$D = \begin{bmatrix} 1 & 1/2 \\ 2 & 1 \end{bmatrix}$$

Then, a decision matrix X $(=[x_{ij}]^T,\ i = 1,2;\ j = 1,2,\ldots 20)$ is designed in terms of the Pareto solutions in Table 9.

$$X = \begin{bmatrix} 104.53 & 184.07 & 93.03 & \cdots & 100.07 & 151.07 & 160.07 \\ 108.46 & 92.65 & 114.96 & \cdots & 110.00 & 94.74 & 93.24 \end{bmatrix}^T$$

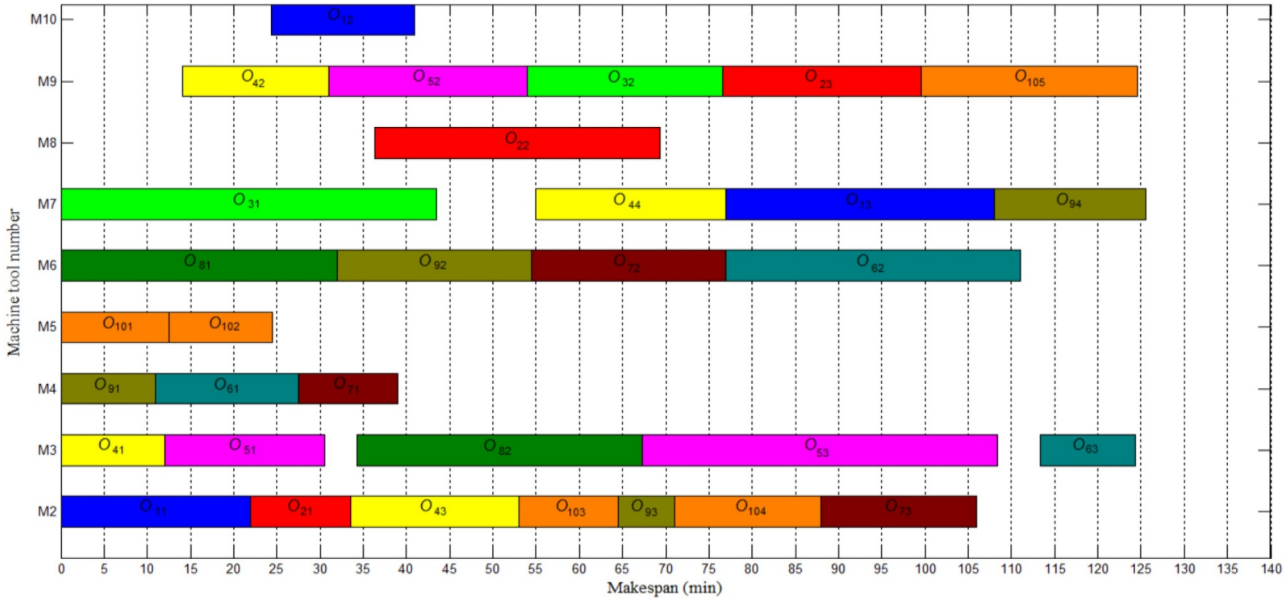


Fig. 9. The most satisfying energy-efficient schedule for the case study.

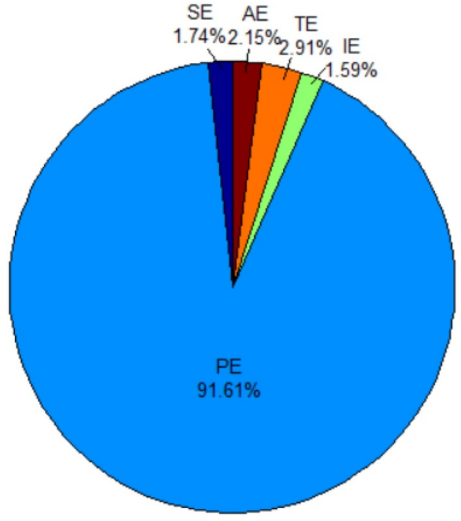


Fig. 10. The pie chart for comparing different energy consumption modules.

Due to different dimensions with respects to the makespan and energy consumption, it is hard to evaluate the performance indicators. By normalizing different criteria values to comparable units, a normalized value y_{ij} is calculated to build a normalized decision matrix Y $(=[y_{ij}]^T,\ i = 1,2;\ j = 1,2,\ldots 20)$.

$$y_{ij} = \frac{\max\{x_{i1}, x_{i2}, \cdots, x_{i20}\} - x_{ij}}{\max\{x_{i1}, x_{i2}, \cdots, x_{i20}\} - \min\{x_{i1}, x_{i2}, \cdots, x_{i20}\}}$$

Hence, the normalized decision matrix can be written as:

$$Y = \begin{bmatrix} 0.9058 & 0.2540 & 1.0000 & \cdots & 0.8281 & 0.7816 & 0.5264 \\ 0.2857 & 0.9807 & 0.0000 & \cdots & 0.5437 & 0.6418 & 0.8677 \end{bmatrix}^T$$

Finally, an evaluation matrix $A = (a_j) = Y \times W$ for the decision-maker satisfaction is given and each element value $a_j (j = 1, 2, \cdots, 20)$ in A is shown in Table 9. It can be observed that $\max\{A\} = a_{17} = 0.7701$, which means that S17 is the most satisfying energy-efficient solution based on the preference of the decision makers.

To evaluate the transportation impact over the objective functions, the most satisfying energy-efficient solution S17, which is called Scheme 1, is analyzed in this paper. Here, the transportation time is taken into account. Note that the transportation time is less than the processing time. The corresponding Gantt chart is depicted in Fig. 9. We can observe that the transportation times before processing operation O_{44}, O_{63}, and O_{82} are integrated into the total idle time of machine tool M_3 and M_7. The transportation could change the makespan and total energy consumption. In addition, we select to shut down M_1 from the perspective of the energy saving, because M_1 has larger rated power and unload power than other machine tools.

With the energy-efficient schedule, the pie chart for comparing the energy consumption in the setup state (*SE*), the energy consumption in the processing state (*PE*), the energy consumption in the idle state (*IE*), the energy consumption in the transportation state (*TE*) and the auxiliary energy consumption in all the states (*AE*) are depicted in Fig. 10. It can be observed that *PE* is crucial to the total energy consumption. Compared to *SE, IE*, and *AE, TE* has more important impact on the total energy consumption.

At the same time, after operations of all jobs are assigned to machine tools, the most satisfying energy-efficient schedule without considering transportation time, which is called Scheme 2, is also analyzed for fair comparison. Table 10 shows the experimental results with regard to Scheme 1 and Scheme 2. Compared to Scheme 2, it is clear that the transportation of Scheme 1 affects both the schedule efficiency and energy efficiency. Considering the transportation time, the makespan increases while the total energy consumption increases. Moreover, the transportation has a significant impact on *IE* and *AE*. For example, for

Table 10
The experimental results with regard to Scheme 1 and Scheme 2.

Schemes	Schedule efficiency Makespan (min)	Energy efficiency *E* (kWh)	*SE* (kWh)	*PE* (kWh)	*IE* (kWh)	*TE* (kWh)	*AE* (kWh)
Scheme 1	125.50	97.03	1.69	88.89	1.54	2.82	2.09
Scheme 2	124.00	94.02	1.69	88.89	1.38	0	2.07

Table 11
The description of three scenarios for different transportation times.

Scenarios	Generation scheme and limit	Description
Scenario 1	Uniform random on (0,5)	The transportation times are lower than the processing times
Scenario 2	Uniform random on (5,40)	The transportation times are close to the processing times
Scenario 3	Uniform random on (40,100)	The transportation times are higher than the processing times

Table 12
Comparison of different energy consumption modules for three scenarios.

Scenarios	Schedule efficiency Makespan (min)	Energy efficiency *E* (kWh)	*SE* (kWh)	*PE* (kWh)	*IE* (kWh)	*TE* (kWh)	*AE* (kWh)
Scenario 1	125.50	97.03	1.69	88.89	1.54	2.82	2.09
Scenario 2	145.00	116.99	1.71	91.53	4.99	16.33	2.42
Scenario 3	182.00	161.41	1.68	96.17	8.50	52.03	3.03

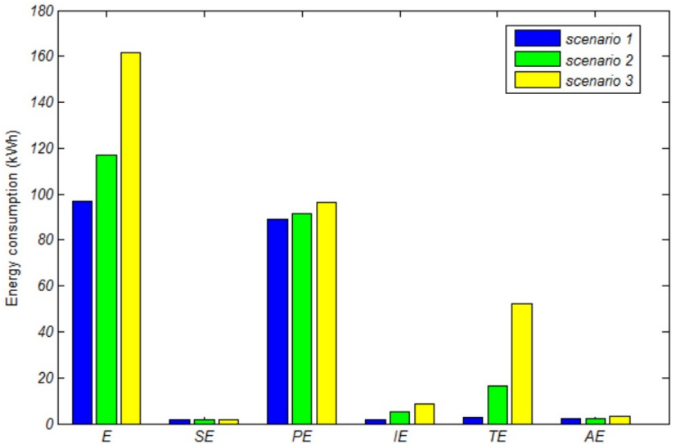


Fig. 11. The histograms of different energy consumption modules for three scenarios.

Table 13
The parameters of the proposed test problems.

Problem size	Number of machine tools	Number of jobs	Number of the total operations
20 × 20	20	20	58
50 × 50	50	50	149
80 × 50	50	80	238

the comparison of *IE* between Scheme 1 and Scheme 2, it has 11.59% idle energy consumption increase for Scheme 1.

Additionally, we consider three scenarios for different levels of transportation times to investigate the effect of transportation on the EFJSP. The transportation times are generated according to processing times of the case study, which follow a uniform distribution [45]. As shown in Table 11, Scenario 1 selects a workshop schedule considering the transportation time that has little impact on the production scheduling and its two objectives; Scenario 2 selects a cross-regional schedule considering the transportation time that has an impact on the production scheduling and its two objectives; Scenario 3 selects a cross-regional schedule considering the transportation time that has a significant impact on the production scheduling and its two objectives. Here, two objectives are the makespan and total energy consumption.

The experimental results are reported in Table 12 and the histograms for comparing *E, SE, PE, IE, TE,* and *AE* are shown in Fig. 11. It is also clear that both the makespan and total energy consumption increase as the transportation time increases, and the ratio of the energy consumption in the transportation state to the total energy consumption becomes higher as the transportation time increases. Furthermore, the increase of the transportation time affects not only *IE, AE,* but also *PE*. In other words, when the transportation times are higher than the processing times, the transportation has a direct impact on the machine tool selection and the operation sequence of each operation for all jobs. And a reasonable facility layout is required for the energy-efficient production scheduling. Therefore, it can be easily concluded that an energy-saving scheduling strategy can be considered by reducing the transportation time for the decision makers.

5.4. Comparison of the EGA and MOGA

In addition, to evaluate the efficiency and validity of the proposed method for investigating the EFJSP, larger scale problems have been considered. We have conducted three test instances: problem 20 × 20, problem 50 × 50, and problem 80 × 50. The parameters of the proposed test problems are listed in Table 13. To analyze the performance of the proposed method for the EFJSP, the proposed EGA is compared with MOGA proposed by May et al. [27]. Here, the comprehensive indicators, HV and IGD, are also used. Due to the stochastic nature of these algorithms, a Wilcoxon's rank sum test is employed to analyze the significant difference between the results generated by the algorithms.

Table 14 shows the results of the larger scale problems with regard to the values of mean and variation coefficient (CV) of HV and IGD, as well as the *h* values of the rank sum test. As a whole, we can observe that the EGA outperforms the MOGA for two comprehensive metrics, HV and IGD, which means that the EGA can obtain the better convergence and distribution of Pareto solutions in the objective space for larger scale EFJSP. In addition, the rank sum test results show that there are significant differences between the EGA and MOGA. The reason is that the guided individuals are well-distributed in the decision space, so it is more likely to search for more Pareto solutions.

Furthermore, the Pareto fronts obtained from one run of each

Table 14
Mean and variation coefficient of two metrics for EGA and MOGA.

Problem size	Algorithms	HV mean(CV) (*h*)	IGD mean(CV) (*h*)
20 × 20	EGA	**2.56(8.85e−02)**	**5.64e−02(9.60e−02)**
	MOGA	2.31(1.49e−01) (1)	1.19e−01(1.48e−01) (1)
50 × 50	EGA	**2.47(1.20e−01)**	**8.07e−02(1.45e−01)**
	MOGA	2.45(1.54e−01) (0)	1.74e−01(2.69e−01) (1)
80 × 50	EGA	**2.58(8.36e−02)**	9.86e−02(2.49e−01)
	MOGA	2.36(2.15e−01) (1)	1.91e−01(2.40e−01) (1)

Table 15
Comparison of the results with EGA and MOGA.

Problem size	MOGA			EGA		
	Pareto solution numbers	Makespan (min)	Energy consumption (kWh)	Pareto solution numbers	Makespan (min)	Energy consumption (kWh)
20 × 20	S1	64	6325.6	S1	63	6173.04
	S2	65	5961.39	S2	64	5507.79
	S3	68	5669.37	S3	66	5037.88
	S4	73	5434.95	S4	69	5029.06
	S5	74	5326.95	S5	70	4848.06
				S6	72	4841.41
				S7	73	4740.89
				S8	74	4669.59
				S9	78	4609.68
				S10	83	4458.03
				S11	86	4436.65
				S12	103	4288.52
50 × 50	S1	85	18,547.12	S1	73	13,048.88
	S2	86	17,069.31	S2	76	12,892.66
	S3	88	16,540.2	S3	77	12,451.6
	S4	92	16,499.81	S4	80	12,141.55
	S5	95	16,427.93	S5	82	11,711.6
	S6	98	16,351.53	S6	84	11,319.33
	S7	111	16,292.73	S7	86	11,149.11
	S8	135	16,019.59	S8	88	11,097.54
	S9	146	15,885.97	S9	89	10,938.72
				S10	90	10,926.17
				S11	97	10,907.02
80 × 50	S1	125	30,363.4	S1	119	21,940.59
	S2	126	30,285.23	S2	120	19,754.49
	S3	128	29,149.66	S3	127	19,712.22
	S4	131	28,947.92	S4	130	19,642.85
	S5	133	28,522.9	S5	135	19,536.97
				S6	136	19,502.74
				S7	137	19,462.18
				S8	162	19,253.63
				S9	165	19,209.25
				S10	171	19,155.6

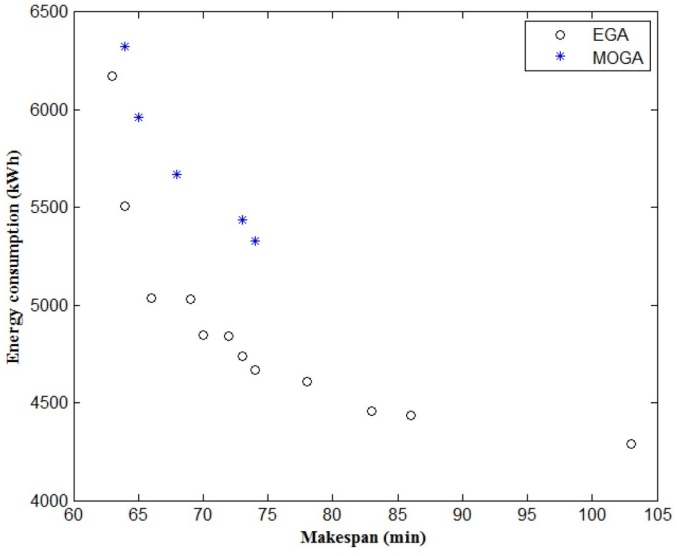


Fig. 12. The Pareto front obtained by EGA versus MOGA for problem 20 × 20.

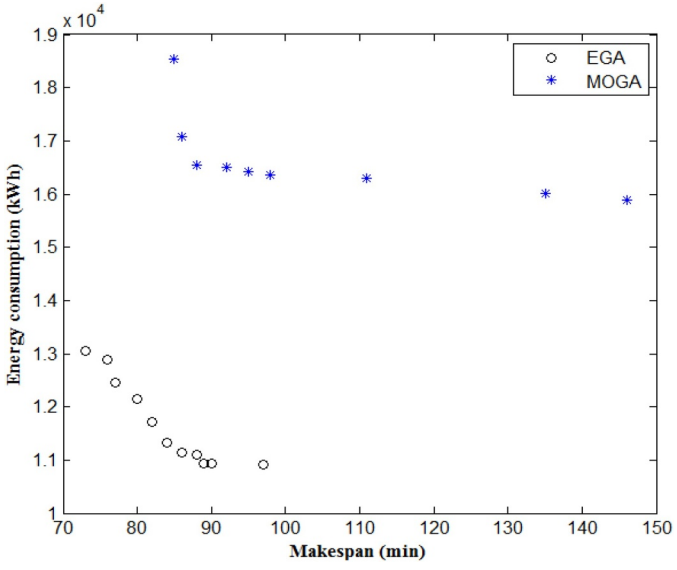


Fig. 13. The Pareto front obtained by EGA versus MOGA for problem 50 × 50.

algorithm are selected to provide an intuitive analysis for these instances as shown in the following tables and figures. Table 15 reports the experimental results of the Pareto solution numbers, makespan and energy consumption. By comparing the Pareto solution number, it is clear that the proposed EGA generates more Pareto solutions than the general MOGA does. It implies that a decision maker has more candidate Pareto solutions to select.

According to Table 15, the comparison results are given in terms of the makespan and energy consumption with the EGA and MOGA for different scale problems as demonstrated in Figs. 12–14. We can observe that the quality of Pareto solutions obtained by EGA is better than the quality of Pareto solutions obtained by MOGA.

6. Conclusions and future work

In this paper, we addressed an energy-efficient flexible job shop scheduling problem with transportation constraints. A multi-objective optimization model is formulated to optimize the total energy

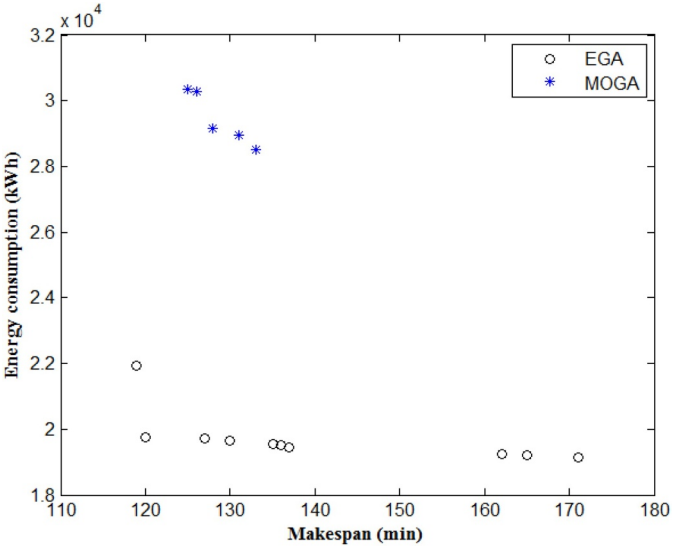


Fig. 14. The Pareto front obtained by EGA versus MOGA for problem 80 × 50.

consumption as well as the traditional makespan objectives. To solve the problem, an enhanced genetic algorithm was developed to obtain the optimized solutions between the makespan and energy consumption. The performance of the proposed EFJSP model and the proposed algorithm were evaluated via comprehensive experiments. The experimental results indicated that the proposed algorithm is effective in solving the EFJSP and it is capable of obtaining better Pareto solutions than the multi-objective genetic algorithm. Furthermore, the results highlighted the conflicting relationship between the makespan and energy consumption, and the transportation energy consumption has a non-negligible impact on the total energy requirement.

In conclusion, the proposed algorithm shows performance improvement compared with other meta-heuristic algorithms, but further studies on the model accuracy and computation time is required through investigating more energy-efficient scheduling problems. On the other hand, unexpected events such as machine failure, rush order and job cancellation, which may occur in practical applications of manufacturing system, should be considered in the EFJSP. Reducing the energy consumption in a dynamic scheduling problem should be also studied in the future.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (NSFC) under Grant no. U1637211, Aeronautical Science Foundation of China under Grant no. 20161652015, the Fundamental Research Funds for the Central Universities under Grant no. NP2017105, Jiangsu Province Qing Lan Project, the Natural Science Foundation of the Jiangsu Higher Education Institutions of China under Grant no. 17KJB460018, the Innovation Foundation for Science and Technology of Yangzhou University under Grant nos. 2016CXJ020 and 2017CXJ018, Science and Technology Project of Yangzhou under Grant no. YZ2017278, Research Topics of Teaching Reform of Yangzhou University under Grant YZUJX2018-28B, and the Spanish Government under the grants TIN2016-80856-R and TIN2015-65515-C4-1-R.

References

[1] EIA, U.S.Energy Information Administration. International Energy Outlook 2016.
[2] Y. Liu, H. Dong, N. Lohse, S. Petrovic, N. Gindy, An investigation into minimising total energy consumption and total weighted tardiness in job shops, J. Clean. Prod. 65 (2014) 87–96.
[3] F. Apostolos, P. Alexios, P. Georgios, S. Panagiotis, C. George, Energy efficiency of manufacturing processes: a critical review, Procedia CIRP 7 (2013) 628–633.
[4] T. Peng, X. Xu, Energy-efficient machining systems: a critical review, Int. J. Adv. Manuf. Technol. 72 (2014) 1389–1406.
[5] L.C. Moreira, W.D. Li, X. Lu, M.E. Fitzpatrick, Energy-Efficient machining process analysis and optimisation based on BS EN24T alloy steel as case studies, Robot. Comput.-Integr. Manuf. 58 (2019) 1–12.
[6] M. Gadaleta, M. Pellicciari, G. Berselli, Optimization of the energy consumption of industrial robots for automatic code generation, Robot. Comput.-Integr. Manuf. 57 (2019) 452–464.
[7] G. Pastras, A. Fysikopoulos, G. Chryssolouris, A theoretical investigation on the potential energy savings by optimization of the robotic motion profiles, Robot. Comput.-Integr. Manuf. 58 (2019) 55–68.
[8] H.E. Nouri, O.B. Driss, K. Ghedira, A classification schema for the job shop scheduling problem with transportation resources: state-of-the-art review, 5th Computer Science On-line Conference, CSOC 2016, April 27, 2016 - April 30, 2016, Springer Verlag, Prague, Czech republic, 2016, pp. 1–11.
[9] C. Gahm, F. Denz, M. Dirr, A. Tuma, Energy-efficient scheduling in manufacturing companies: a review and research framework, Eur. J. Oper. Res. 248 (2016) 744–757.
[10] A. Giret, D. Trentesaux, V. Prabhu, Sustainability in manufacturing operations scheduling: a state of the art review, J. Manuf. Syst. 37 (Part 1) (2015) 126–140.
[11] M. Akbar, T. Irohara, Scheduling for sustainable manufacturing: a review, J. Clean. Prod. 205 (2018) 866–883.
[12] G. Mouzon, M.B. Yildirim, J. Twomey, Operational methods for minimization of energy consumption of manufacturing equipment, Int. J. Prod. Res. 45 (2007) 4247–4271.
[13] G. Mouzon, M.B. Yildirim, A framework to minimise total energy consumption and total tardiness on a single machine, Int. J. Sustain. Eng. 1 (2008) 105–116.
[14] M.B. Yildirim, G. Mouzon, Single-machine sustainable production planning to minimize total energy consumption and total completion time using a multiple objective genetic algorithm, IEEE Trans. Eng. Manag. 59 (2012) 585–597.
[15] F. Shrouf, J. Ordieres-Meré, A. García-Sánchez, M. Ortega-Mier, Optimizing the production scheduling of a single machine to minimize total energy consumption costs, J. Clean. Prod. 67 (2014) 197–207.
[16] A. Che, X. Wu, J. Peng, P. Yan, Energy-efficient bi-objective single-machine scheduling with power-down mechanism, Comput. Oper. Res. 85 (2017) 172–183.
[17] K. Fang, N. Uhan, F. Zhao, J.W. Sutherland, A new approach to scheduling in manufacturing for power consumption and carbon footprint reduction, J. Manuf. Syst. 30 (2011) 234–240.
[18] K. Fang, N.A. Uhan, F. Zhao, J.W. Sutherland, Flow shop scheduling with peak power consumption constraints, Ann. Oper. Res. 206 (2013) 115–145.
[19] J.-Y. Ding, S. Song, C. Wu, Carbon-efficient scheduling of flow shops by multi-objective optimization, Eur. J. Oper. Res. 248 (2016) 758–771.
[20] S.A. Mansouri, E. Aktas, U. Besikci, Green scheduling of a two-machine flowshop: trade-off between makespan and energy consumption, Eur. J. Oper. Res. 248 (2016) 772–788.
[21] C. Lu, L. Gao, X. Li, Q. Pan, Q. Wang, Energy-efficient permutation flow shop scheduling problem using a hybrid multi-objective backtracking search algorithm, J. Clean. Prod. 144 (2017) 228–238.
[22] A.A.G. Bruzzone, D. Anghinolfi, M. Paolucci, F. Tonelli, Energy-aware scheduling for improving manufacturing process sustainability: a mathematical model for flexible flow shops, CIRP Ann. – Manuf. Technol. 61 (2012) 459–462.
[23] H. Luo, B. Du, G.Q. Huang, H. Chen, X. Li, Hybrid flow shop scheduling considering machine electricity consumption cost, Int. J. Product. Econ. 146 (2013) 423–439.
[24] J. Yan, L. Li, F. Zhao, F. Zhang, Q. Zhao, A multi-level optimization approach for energy-efficient flexible flow shop scheduling, J. Clean. Prod. 137 (2016) 1543–1552.
[25] M. Dai, D. Tang, A. Giret, M.A. Salido, W.D. Li, Energy-efficient scheduling for a flexible flow shop using an improved genetic-simulated annealing algorithm, Robot. Comput.-Integr. Manuf. 29 (2013) 418–429.
[26] Y. Liu, H. Dong, N. Lohse, S. Petrovic, A multi-objective genetic algorithm for optimisation of energy consumption and shop floor production performance, Int. J. Prod. Econ. 179 (2016) 259–272.
[27] G.K. May, B. Stahl, M. Taisch, V. Prabhu, Multi-objective genetic algorithm for energy-efficient job shop scheduling, Int. J. Prod. Res. 53 (2015) 7071–7089.
[28] R. Zhang, R. Chiong, Solving the energy-efficient job shop scheduling problem: a multi-objective genetic algorithm with enhanced local search for minimizing the total weighted tardiness and total energy consumption, J. Clean. Prod. 112 (2016) 3361–3375.
[29] M.A. Salido, J. Escamilla, A. Giret, F. Barber, A genetic algorithm for energy-efficiency in job-shop scheduling, Int. J. Adv. Manuf. Technol. 85 (2016) 1303–1314.
[30] H. Mokhtari, A. Hasani, An energy-efficient multi-objective optimization for flexible job-shop scheduling problem, Comput. Chem. Eng. 104 (2017) 339–352.
[31] Q. Liu, M. Zhan, F.O. Chekem, X. Shao, B. Ying, J.W. Sutherland, A hybrid fruit fly algorithm for solving flexible job-shop scheduling to reduce manufacturing carbon footprint, J. Clean. Prod. 168 (2017) 668–678.
[32] L. Yin, X. Li, L. Gao, C. Lu, Z. Zhang, A novel mathematical model and multi-objective method for the low-carbon flexible job shop scheduling problem, Sustain. Comput. 13 (2017) 15–30.
[33] L. Zhang, Q. Tang, Z. Wu, F. Wang, Mathematical modeling and evolutionary generation of rule sets for energy-efficient flexible job shops, Energy 138 (2017) 210–227.
[34] X. Wu, Y. Sun, A green scheduling algorithm for flexible job shop with energy-saving measures, J. Clean. Prod. 172 (2018) 3249–3264.
[35] H. Wang, Z. Jiang, Y. Wang, H. Zhang, Y. Wang, A two-stage optimization method for energy-saving flexible job-shop scheduling based on energy dynamic characterization, J. Clean. Prod. 188 (2018) 575–588.
[36] G. Gong, Q. Deng, X. Gong, W. Liu, Q. Ren, A new double flexible job-shop

scheduling problem integrating processing time, green production, and human factor indicators, J. Clean. Prod. 174 (2018) 560–576.
[37] A. Rossi, G. Dini, Flexible job-shop scheduling with routing flexibility and separable setup times using ant colony optimisation method, Robot. Comput.-Integr. Manuf. 23 (2007) 503–516.
[38] Q. Zhang, H. Manier, M.-A. Manier, A modified shifting bottleneck heuristic and disjunctive graph for job shop scheduling problems with transportation constraints, Int. J. Prod. Res. 52 (2014) 985–1002.
[39] C.-D. Liou, Y.-C. Hsieh, A hybrid algorithm for the multi-stage flow shop group scheduling with sequence-dependent setup and transportation times, Int. J. Prod. Econ. 170 (2015) 258–267.
[40] S. Karimi, Z. Ardalan, B. Naderi, M. Mohammadi, Scheduling flexible job-shops with transportation times: mathematical models and a hybrid imperialist competitive algorithm, Appl. Math. Model. 41 (2017) 667–682.
[41] Y. He, Y. Li, T. Wu, J.W. Sutherland, An energy-responsive optimization method for machine tool selection and operation sequence in flexible machining job shops, J. Clean. Prod. 87 (2015) 245–254.
[42] F. Liu, J. Xie, S. Liu, A method for predicting the energy consumption of the main driving system of a machine tool in a machining process, J. Clean. Prod. 105 (2015) 171–177.
[43] G. Zhang, X. Shao, P. Li, L. Gao, An effective hybrid particle swarm optimization algorithm for multi-objective flexible job-shop scheduling problem, Comput. Ind. Eng. 56 (2009) 1309–1318.
[44] T.L. Saaty, How to make a decision: the analytic hierarchy process, Eur. J. Oper. Res. 24 (1994) 19–43.
[45] C.D. Liou, Y.C. Hsieh, A hybrid algorithm for the multi-stage flow shop group scheduling with sequence-dependent setup and transportation times, Int. J. Prod. Econ. 170 (2015) 258–267.